

Shane Paton

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Clearance: Interim Secret (TS/SCI in progress)

Education

Bachelor of Science in Software Engineering
Miami University, Oxford, OH
GPA: 3.97 / 4

Expected 05/2027

Skills

Languages: C++, Java, Dart, JavaScript, Python, HTML, CSS, SQL

Technologies & Concepts: React, Flutter, MIXR, Firebase, Git, GitHub, Jira, Data Structures and Algorithms

Work Experience

Modern Technology Solutions, Inc. (MTSI) - Software Engineering Intern 05/2025 – 08/2025

- Achieved synchronized simulations across 5+ networked devices by leading networking implementation with Distributed Interactive Simulation (DIS) protocols over UDP, ensuring fast and reliable network communication
- Simulated realistic object behavior with a custom Plinko physics engine supporting 10+ balls at 60 FPS
- Reduced file size by up to 60% with a custom Huffman compressor/decompressor in C++, visualized decompressing and the Huffman Tree using MIXR

Noshable - Website Developer Intern 05/2024 – 08/2024

- Enhanced data visualization by building a dynamic map page using Google Maps API, displaying over 9,000 custom markers to help efficiently organize cleaner availability and client onboarding
- Utilized Firebase for cloud storage to streamline datasets and reduce unnecessary API calls, saving ~\$2,000 monthly
- Revamped UX by building interactive UI components in Flutter simplifying navigation and usability

Projects

Spotify Song Ranker - React

- Introduced song searching via Spotify API, enabling users to fetch tracks, streamlining user exploration & interaction
- Incorporated real-time UI updates by managing state with React hooks, improving responsiveness & user interaction
- Implemented drag-and-drop song ranking with React, replicating Spotify's design for a familiar and intuitive UX

Order Matching Engine – C++

- Designed a multithreaded, low-latency order matching engine using producer-consumer queues, capable of processing 750,000+ orders per second under load
- Created a custom memory pool cutting dynamic memory allocation by over 50% and boosting performance
- Built scalable batching and synchronization mechanisms ensuring thread safety and efficient resource utilization

Marathon Map - Flutter

- Enabled scalable cloud synchronization handling 1,000+ user entries in testing without sync issues by integrating Firebase Firestore, securely tracking and storing user data in real time
- Built dynamic forms for all 50 U.S. states, updating a map view in real time to enhance data visualization
- Constructed an editable log page allowing users to manipulate entries, increasing app usability

Campus Involvements

Sports Modeling & Analytics Club - Co-founder & CTO 05/2025 - Present

- Boosted personalized user engagement by implementing authentication, role-based access control, and real-time data sync with React, Next.js, and Supabase, allowing users to create bets and posts and admins to control content
- Launching a full-stack web application leveraging Cursor and other AI tools to support ~50 users in managing sports analytics and betting insights
- Learned AI modeling techniques aiming to develop a predictive college football betting model leveraging machine learning and statistical analysis

Association for Computing Machinery (ACM) - Member 02/2025 - Present

- Discussed trending topics in computing and software development, including AI agents and web scrapers

Techstars Startup Weekend 08/2023

- Collaborated with a team of five to construct a prototype and pitch for a metal awareness clothing brand, delivering a mock VC presentation to an audience of ~100 people
- Generated \$1,000 in revenue over and placed third winning another \$1,000 as a team, demonstrating entrepreneurial mindset, rapid prototyping, and effective teamwork